

# Mammalia Raccoon Proximity Network Analysis

<https://github.com/aiyshawy/Raccoon-Network-Analysis>

Python

NetworkX

Jupyter

Graph Analysis

Epidemiology

## OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

## IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

## TECHNICAL WALKTHROUGH

### Mammalia Raccoon Proximity Network Analysis – Project

#### Explanation

##### Project Overview

This project analyzes social interaction patterns among urban raccoons using network analysis to understand how rabies and other pathogens can spread within raccoon populations.

Since raccoons are a major vector of rabies, studying how often and how closely they interact helps estimate disease transmission risk and supports better public health and wildlife management strategies.

#### Objective

##### Model raccoon interactions as a graph/network

Identify high-risk raccoons that can spread pathogens quickly Use centrality measures to understand social structure and disease flow

##### Dataset & Network Construction

24 raccoons → represented as nodes

~2000 interactions → represented as weighted edges

##### Data collected using proximity-detecting collars

Edge weight = time spent in close proximity Monthly interaction data used to build the network

**If two raccoons are connected, it means they had close physical interaction, which is critical for rabies transmission.**

Why Network Analysis?

Disease spread is not random.

#### It depends on

i) Who interacts with whom

ii) How frequently

iii) Who acts as a bridge between groups

iv) Network analysis helps identify super-spreaders and critical connectors.

## Centrality Measures Used & Interpretation

### Degree Centrality

i) Measures number of direct connections

ii) High degree = more interactions

### Inference

i) Raccoon #4 has the highest degree

ii) Acts as a high-contact individual

iii) High risk of initiating widespread infection

### Closeness Centrality

Measures how close a node is to all others

#### Indicates how fast something can spread

Raccoon #2 has highest closeness centrality

Can spread pathogens rapidly across the entire network

### Betweenness Centrality

Measures how often a node lies on shortest paths

#### Identifies bridge nodes

Raccoon #16 connects different subgroups

#### Acts as a key transmission bridge

Removing or vaccinating this raccoon can significantly slow disease spread

### Clustering Coefficient

Measures how tightly a raccoon is embedded in a local group

#### Indicates family or group behavior

Raccoon #2 has highest clustering

Infection here may cause localized outbreaks within families

### Eigenvector Centrality

Measures influence based on connections to influential nodes

Raccoon #4 again ranks highest

**Connected to other highly interactive raccoons**

**Considered a super-spreader candidate**

### **PageRank Algorithm**

Evaluates overall importance considering direction and weight of interactions

**Helps rank raccoons by global influence**

### **Key Findings**

**Raccoon populations are highly interconnected**

**Disease can spread much faster than expected**

A few raccoons play disproportionately large roles in pathogen transmission

Targeted intervention (vaccination/tracking of key raccoons) is more effective than random control

### **Technologies & Concepts Used**

**Graph Theory**

**Weighted graphs**

**Centrality algorithms**

**Epidemiological modeling concepts**

### **KEY DETAILS**

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1. Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.
2. Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.
3. Identified superspreader nodes whose removal would most disrupt disease propagation chains.
4. Visualized network topology and centrality distributions to communicate findings clearly.
5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.
6. Analyzed a dense 24-node, 226-edge proximity graph (~82% density) derived from 1,997 weighted temporal contact records.
7. Computed clustering coefficients and aggregated per-node contact strength (total edge weight) to quantify interaction intensity — node 4 had the highest total contact weight (956,341).

8. Rendered the network using Kamada-Kawai force-directed layout (Matplotlib) to reveal spatial clustering patterns in the proximity data.